

Topology of the real line – 1

Open and closed sets, closure and boundary, density

Homework

8 сентября 2026 года

Theoretical minimum. Interior, closure and boundary and the duality $\text{int}(E^c) = (\overline{E})^c$; the structure theorem for open subsets of $\mathbb{R}$; dense and nowhere dense sets and the three equivalent descriptions of nowhere density; F_σ and G_δ sets; the standard counting device — a disjoint family of intervals is at most countable because each member contains a rational point.

1. Prove the three characterisations of nowhere density listed in Theorem 2, and show that a closed nowhere dense subset of $[0, 1]$ may have Lebesgue measure arbitrarily close to 1; such a set is built in the next sheet.
2. Prove that every closed subset of $\mathbb{R}$ is a G_δ and every open subset is an F_σ . Deduce that every closed set is the zero set of a continuous function, and that the distance function $x \mapsto \text{dist}(x, F)$ realises it.
3. (IMC) Prove or disprove: (a) there exists a monotone $f: [0, 1] \rightarrow [0, 1]$ such that for each $y \in [0, 1]$ the equation $f(x) = y$ has uncountably many solutions; (b) there exists a continuously differentiable $f: [0, 1] \rightarrow [0, 1]$ with the same property.
4. (Karasev 1.52) Can the interval (a, b) be written as a union of countably many pairwise disjoint closed segments? A single point counts as a segment. *Hint:* look at the boundary of the union of the segments already used.
5. (Putnam 1994) Let $(r_n)_{n \geq 0}$ be positive reals with $r_n \rightarrow 0$, and let S be the set of numbers representable as $r_{i_1} + r_{i_2} + \dots + r_{i_{1994}}$ with $i_1 < i_2 < \dots < i_{1994}$. Prove that every non-empty interval (a, b) contains a non-empty subinterval (c, d) disjoint from S ; that is, S is nowhere dense.

6. (IMC) Let S be an infinite set of real numbers such that $|s_1 + s_2 + \cdots + s_k| < 1$ for every finite subset $\{s_1, \dots, s_k\} \subseteq S$. Show that S is countable.
7. (Putnam 1989) Can a countably infinite set have an uncountable collection of non-empty subsets such that the intersection of any two of them is finite? Answer the same question with “finite” replaced by “of at most 1989 elements”.
8. (IMC) Let $\mathcal{S}$ be a family of spheres in $\mathbb{R}^n$, $n \geq 2$, such that the intersection of any two of them contains at most one point. Prove that the set of points which belong to at least two different spheres of $\mathcal{S}$ is countable.
9. Construct a set $A \subseteq \mathbb{R}$ such that A and its complement are both dense in every interval, and compute ∂A . Then construct a set whose boundary is all of $\mathbb{R}$ and one whose boundary is a single point.
10. (Karasev 7.4) Suppose a family $\mathcal{C}$ of subsets of $\mathbb{N}$ is such that for any $X, Y \in \mathcal{C}$ either $X \subseteq Y$ or $Y \subseteq X$. Must $\mathcal{C}$ be at most countable? Compare the answer with the previous problem: both are about how large an almost linearly ordered family of subsets of a countable set can be.